

The Role of Color in Pedagogical and Learning Environments and Children's Perception

Version 1.0

Initial Conceptual Research Preprint

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ABSTRACT

The research opens an initial conceptual investigation into the role of color in pedagogical and learning environments, with particular attention to children's perceptual experiences and the possibilities introduced by generative artificial intelligence in the creation and modification of visual environments.

The starting point of the investigation is the perception and interpretation of color within an already existing visual environment. With the emergence of generative artificial intelligence, however, visual environments and their color configurations can also be created and modified within a short period of time during the generation process. This raises new questions concerning the relationship between the creation of an image, the configuration of color, and its perceptual processing.

The central question of this initial investigation is how the role of color in pedagogical and learning environments can be interpreted when visual environments can be dynamically created and modified with the help of generative artificial intelligence.

The present Version 1.0 does not seek to present a completed empirical investigation or definitive results. Its aim is to establish the research problem, define the initial conceptual framework of the investigation, and open a research direction that examines the relationship between color, children's perception, and AI-generated visual environments.

This Version 1.0 documents the initial state of the research and seeks to provide a foundation for subsequent theoretical, visual, and empirical stages of the research.

KEYWORDS

Color perception; Children; Learning environments; Pedagogical environments; Generative artificial intelligence; Visual environments; Color variables; Visual perception; Educational design; Generative image systems

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INTRODUCTION

The perception and recognition of color are fundamental parts of visual experience. In pedagogical and learning environments, the child encounters colors as part of an already created visual environment: they perceive, distinguish, and interpret them in the context of the forms and spaces appearing around them.

In this situation, the creation of the visual environment and its subsequent perception can be considered as two distinct stages. The creation of an image or space is preceded by a process of design and execution, the result of which the child later enters into a perceptual relationship with.

The emergence of generative artificial intelligence creates a new situation. AI-based image generation systems make it possible for a visual environment to be created within a short period of time based on a natural language instruction, and then modified through further instructions. In this process, color can become an explicitly specified or modified element of the visual environment, rather than only a property of a previously established visual configuration.

This technological change raises new questions concerning the relationship between color and perception. Compared to the perception of a previously created visual environment, an additional generative process emerges, which also makes the creation of the visual environment itself dynamic.

As a consequence, the question arises as to what significance color may have in children's perceptual experience when the visual environments surrounding them or presented to them do not necessarily exist in a predetermined form, but can be dynamically created and modified with the help of generative artificial intelligence.

The research initiates the investigation from this point: by exploring the new relationship between the perception of color and visual environments created or modified through generative artificial intelligence.

1. COLOR AS A GENERATIVE VARIABLE

In this research, color is not to be understood as an independent, isolated visual property. Its meaning develops within the complete visual structure of the image, in relation to forms, spatial relationships, contrasts, lighting conditions, and surrounding colors. Therefore, the perceptual significance of the same color may differ depending on the visual environment in which it appears.

One of the characteristics of generative image systems is that visual components do not necessarily come together through a fixed sequence of independently controlled parameters. During the creation of an image, color, form, composition, lighting, and other visual attributes may be generated in relation to one another. Therefore, modifying a color-related instruction does not necessarily mean modifying only a single visual element, but may also result in changes to the wider visual configuration.

This relationship is important for the present research because it introduces a distinction between color as a visual property and color as a variable within the process of generating a visual environment. In the first case, color is observed as part of an already existing image or space. In the second, color may be intentionally specified, modified, or varied during the creation of the visual environment. The research uses the term “generative variable” in this latter sense: not as a claim that color can always be independently controlled by a generative system, but as a visual property that can become a deliberate subject of variation within the generation process.

It follows that the study of color cannot be limited to determining whether a child recognizes a particular color. From the perspective of the research, it becomes more important to examine how color relates to the other visual elements of the environment, what differences can be perceived between different color configurations, and how the perceptual structure of the visual environment changes when its color configuration is modified.

From this perspective, the generative environment also raises a methodological possibility. Multiple versions of a visual situation can be generated in which color configurations are intentionally varied. Comparing these versions may provide an opportunity to examine the

perceptual consequences of different color configurations and to investigate the extent to which these consequences can be attributed to color itself or to changes occurring simultaneously in the wider visual configuration.

This distinction introduces an important methodological question for later stages of the research: to what extent can color be varied while other relevant visual properties remain sufficiently controlled? If a change in color is accompanied by changes in composition, lighting, form, texture, or other visual characteristics, the resulting perceptual response cannot necessarily be attributed to color alone. The ability to control and compare these variables will therefore become an important part of the subsequent research design.

Therefore, in later stages of the research, study situations can be developed in which color appears as a variable controlled in different ways. For example, hue, saturation or chroma, lightness, luminance contrast, the combination of colors, color variability, and the spatial distribution of color within the visual environment may become subjects of investigation. Their relationship with other elements of the visual environment may also be examined in order to determine whether particular perceptual effects are associated with color itself or with the broader configuration in which color appears.

In this approach, generative artificial intelligence is not itself the subject of the research, but rather a tool that may enable the rapid creation and modification of different visual configurations. Its relevance to the research lies in the possibility of generating multiple visual situations in which selected aspects of color can be intentionally varied and compared. The focus of the research remains the relationship between color, visual configuration, and children's perceptual experience.

Based on the approach outlined in this chapter, one possible direction for further research is to examine how generated color variations can be used to create comparable visual situations, how reliably the relevant visual properties can be controlled, and what perceptual differences can be observed when children encounter these different configurations.

CONCLUSION

Based on the present stage of the research, color is approached not only as an object of recognition or identification, but as a component of the broader visual environment in which perception takes place. Its perceptual significance may depend on its relationship to form, spatial organization, contrast, lighting, surrounding colors, and other elements of the visual configuration.

The introduction of generative artificial intelligence adds a further dimension to this investigation. Visual environments can be created, modified, and varied through a generative process, making it possible to consider color not only as an observed property of an existing environment, but also as a factor that can be intentionally varied during the creation of visual stimuli. At the same time, this possibility introduces an important methodological question: whether changes in color can be sufficiently controlled and distinguished from simultaneous changes in other visual properties.

This distinction is central to the further development of the research. If different color configurations are to be compared in relation to children's perceptual experience, it will be necessary to examine not only what visual differences are produced, but also how reliably those differences can be attributed to color rather than to changes in composition, lighting, form, texture, or overall visual complexity.

Therefore, the principal outcome of Version 1.0 is not an empirical finding, but the formulation of a research problem and an initial conceptual framework through which that problem can be further investigated. The research proposes that the relationship between color and children's visual experience may be examined through controlled variations of visual environments, while recognizing that the degree of control achievable within generative systems remains itself an issue for investigation.

At this stage, the research remains open and exploratory. Subsequent versions may refine the conceptual definition of color as a research variable, examine the possibilities and limitations of generative systems for creating comparable visual situations, and develop appropriate theoretical, visual, and empirical approaches for investigating children's perceptual responses.

The development from this initial conceptual stage toward empirical investigation will therefore depend on establishing sufficiently clear relationships between the visual variables being manipulated, the characteristics of the generated environments, and the perceptual responses that may be observed. Version 1.0 establishes this starting point without presuming the outcome of the subsequent investigation.

RESEARCH ETHOS STATEMENT

This research is presented as an independent theoretical and conceptual inquiry. Version 1.0 does not present an empirical study or claim empirically established effects of color on children's perception or learning. The propositions presented here are intended as a preliminary conceptual framework for subsequent investigation.

The research distinguishes between observations, conceptual propositions, and findings that would require empirical verification. Accordingly, the present version does not assume that particular colors produce specific perceptual, psychological, or educational effects, nor does it treat the possibilities of generative artificial intelligence as established methodological capabilities.

The purpose of Version 1.0 is to establish the research problem and its initial conceptual boundaries while leaving its empirical development open. Subsequent research may confirm, refine, qualify, or reject the propositions introduced in this version.

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